A FINITE TYPE MORPHISM BETWEEN REGULAR SCHEMES IS A LOCAL COMPLETE INTERSECTION MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18337-18342 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18344-18351, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Frequently quoted structural fact; its two-reference proof is complete at the stated precision.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Algebra, lemma regular-goes-up; Divisors, lemma immersion-regular-regular-immersion; the definition of local complete intersection morphism via local factorization into a regular immersion followed by a smooth morphism (chapter definition section).

Source reference. Original lemma 07NF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18351–18351. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E9J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18351–18351. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. If f is locally of finite type and X and Y are regular, then f is a local complete intersection morphism.*

Proof. We may assume there is a factorization $X \to \mathbf{A}^n_Y \to Y$ where the first arrow is an immersion. As Y is regular also $\mathbf{A}^n_Y$ is regular by Algebra, Lemma 07NF. Hence $X \to \mathbf{A}^n_Y$ is a regular immersion by Divisors, Lemma 0E9J. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.